Experiment Set 4

Submission Due Date: 20/11/2026

Experiment 41

Set-up of Raspberry Pi for the experiments:

Perform the required steps to initialize Raspberry Pi and complete the set-up of Raspberry Pi.

Answer Following Questions in your report..

Question 41.a: Describe the steps involved in completing the set-up of Raspberry Pi in detail.

Experiment 42

Write a Python Program for Raspberry Pi that performs following task.

- Blink one LED connected to Pin No. 11. The LED should be ON for one second and OFF for one second.
- Connect the hardware circuit to perform and implement above program. Use a series resistor with LED while implementing the circuit.

Answer Following Questions in your report..

Question 42.a: Prepare a video recording with appropriate voice over explaining the execution of this experiment. Save the video in any folder/drive and include the link to that video in this report.

Question 42.b: Include the code

Experiment 43

Modify the program in Experiment No. 42 to generate 20% duty cycle for LED to remain turn ON.

That means, the LED will be OFF for 80% of time and ON for 20% of time.

Answer Following Questions in your report..

Question 43.a: Prepare a video recording with appropriate voice over explaining the execution of this experiment. Save the video in any folder/drive and include the link to that video in this report.

Question 43.b: Include your new Python Program in your report.

Question 43.c: Describe the changes you have made in the program.

Experiment 44

Modify and update hardware, without changing the program of experiment No. 43, such that two LEDs behave in the following way.

LED 1: will be OFF for 80% of time and ON for 20% of time.

LED 2: will be OFF for 20% of time and ON for 80% of time.

Answer Following Questions in your report..

Question 44.a: Prepare a video recording with appropriate voice over explaining the execution of this experiment. Save the video in any folder/drive and include the link to that video in this report.

Question 44.b: Describe the changes you have made in the hardware connections.

Experiment 45

Write a Python program for Raspberry Pi to blink LEDs on Pins 11 , 13, and 15 simultaneously every second.

Execute the program and observe the output on hardware.

Answer Following Questions in your report..

Question 45.a: Prepare a video recording with appropriate voice over explaining the execution of this experiment. Save the video in any folder/drive and include the link to that video in this report.

Question 45.b: Include your new Python Program in your report.

Experiment 46

Write a Python program for Raspberry Pi to -

(1) to turn on LED on Pins 11 , 13, and 15 one after another with delay of 2 second. (This means, first LED on pin 11 turns on. After 2 second delay, second LED on pin 13 turns on. This continues till all three LEDs turn ON)

(2) After all three LEDs are on, turn all of them OFF together.

(3) Repeat above steps continuously.

Execute the program and observe the output on hardware.

Answer Following Questions in your report..

Question 46.a: Prepare a video recording with appropriate voice over explaining the execution of this experiment. Save the video in any folder/drive and include the link to that video in this report.

Question 46.b: Include your new Python Program in your report.

Experiment 47

Assume three LEDs are connected to Pins 11, 13, and 15 of Raspberry Pi. Write a python program to turn on one LED at a time, in the following way.

Second 1: LED on Pin 11 turns on, all other LEDs are off
Second 2: LED on Pin 11 turns off, LED on pin 13 turns on, other LEDs are off.
Second 3: LED on Pin 13 turns off, LED on pin 15 turns on, other LEDs are off
Second 4: LED on Pin 15 turns off, LED on pin 11 turns on, other LEDs are off.

Use delay of 2 Seconds between each transactions.

Above cycle repeats continuously.

Execute the program and observe the output on hardware.

Save this new program.

Answer Following Questions in your report..

Question 47.a: Prepare a video recording with appropriate voice over explaining the execution of this experiment. Save the video in any folder/drive and include the link to that video in this report.

Question 47.b: Include your new Python Program in your report.

Experiment 48

Assume three LEDs are connected to Pins 11, 13, and 15 of Raspberry Pi. Assume that two press button switches are connected to Pins 29 and 31. Connect the switches such that if the switch is pressed, Logic 1 goes to Raspberry Pi. Write a Python program that performs following operations.

If switch on Pin 29 is pressed, turn on only LED connected to Pin 11.

If switch on Pin 31 is pressed, turn on two LEDs connected to Pins 13 and 15.

Answer Following Questions in your report..

Question 48.a: Prepare a video recording with appropriate voice over explaining the execution of this experiment. Save the video in any folder/drive and include the link to that video in this report.

Question 48.b: Include your new Python Program in your report.

Experiment 49

Modify the program in Experiment 48 such that, when no switch is pressed, all LEDs are ON.

If switch on Pin 29 is pressed, turn OFF only LED connected to Pin 11.

If switch on Pin 31 is pressed, turn OFF two LEDs connected to Pins 13 and 15.

Answer Following Questions in your report..

Question 49.a: Prepare a video recording with appropriate voice over explaining the execution of this experiment. Save the video in any folder/drive and include the link to that video in this report.

Question 49.b: Include your new Python Program in your report.

Question 49.c: Describe the changes you have made in the program.

Experiment 50

1. Connect the Ultrasonic Sensor to Raspberry Pi.

2. Continuously send the data from the Ultrasonic Sensor to the website “ThingSpeak.com”. Observe the data and Graph.

Answer Following Questions in your report..

Question 50.a: Prepare a video recording with appropriate voice over explaining the execution of this experiment. Save the video in any folder/drive and include the link to that video in this report.

Question 50.b: Include your new Python Program in your report.

Experiment 51

1. Connect the DHT22 (Temperature and Humidity Sensor) to Raspberry Pi.

2. Continuously send the data from the DHT22 to the website “ThingSpeak.com”. Observe the data and Graph.

Answer Following Questions in your report..

Question 51.a: Prepare a video recording with appropriate voice over explaining the execution of this experiment. Save the video in any folder/drive and include the link to that video in this report.

Question 51.b: Include your new Python Program in your report.

Experiment 52

1. Connect the IR Sensor to Raspberry Pi.

2. Continuously send the data from the IR Sensor to the website “ThingSpeak.com”. Observe the data and Graph.

Answer Following Questions in your report..

Question 52.a: Prepare a video recording with appropriate voice over explaining the execution of this experiment. Save the video in any folder/drive and include the link to that video in this report.

Question 52.b: Include your new Python Program in your report.